

Introduction to Machine Learning

Machine Learning (ML) is a subset of Artificial Intelligence (AI) that focuses on building systems that learn from data, improve their performance over time, and make predictions or decisions without being explicitly programmed for every task.

1. Types of Machine Learning

Supervised Learning: The model is trained on labeled data. The goal is to learn a mapping function from input to output. Example: $y = f(x)$.

Unsupervised Learning: The model works with unlabeled data to find hidden patterns or structures. Example: Clustering algorithms.

Reinforcement Learning: An agent learns to make decisions by performing actions in an environment to maximize a reward signal.

2. The Core Pipeline

The ML process generally involves: Data Collection → Data Preprocessing → Feature Engineering → Model Selection → Training → Evaluation → Deployment.

3. Key Mathematical Concepts

At its core, ML relies on optimization. A common objective is minimizing a Loss Function L :

$$L(\theta) = \sum (y_i - f(x_i; \theta))^2$$

Where θ represents the model parameters being optimized via techniques like Gradient Descent.